# Neuroanatomical Mapping of Upstream Targets of Somatostatin in the Dorsal Motor Nucleus and Nucleus of the Solitary Tract by Retrograde Uptake of Cre-Dependent Pseudorabies Virus

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**Abstract**

Based on previous studies, we know anorexigenic agonists exert inhibitory control over somatostatin-gamma-Aminobutyric acid (Sst-GABA) neurons in the dorsal motor nucleus of the vagus (DMV) and nucleus of the solitary tract (NTS). We aimed to map these efferent regions of the vagal-gastric circuit that may regulate Sst-GABA neurons and, in turn, gastric function and motility in the stomach. We used the Cre-dependent pseudorabies virus (PRV)-Introvert as a neuroanatomical tracer. Following injection into various regions of the stomach of transgenic Sst-Cre mice, this polysynaptic virus travelled upstream into the brain via retrograde uptake and, upon interaction with Cre-expressing Sst neurons, activated and labelled upstream targets with enhanced green fluorescent protein (EGFP). We then analyzed the confocal images of harvested brain slices using ImageJ and overlay software to determine which brain regions act as a source of anorexigenic agonists on Sst neurons in the DMV and NTS. We found strong evidence of an anatomical connection between the gastric antrum and the hypothalamus and amygdala through the vagal-vago circuit.

## Section I: Background and Introduction

### *The Role of Sst-GABA Neurons in Gastric Function*

The dorsal motor nucleus of the vagus (DMV) plays an important role in the regulation of gastric function (Chan, Ko, & Cho, 1995), especially through local gamma-Aminobutyric acid (GABA) signalling across preganglionic neurons projecting to the stomach (Cruz et al., 2019).

In a previous study, Lewin et al (2016) established that somatostatin (Sst)-GABA neurons are anatomically and functionally connected to the DMV-output neurons that innervate the gastric antrum. They established an anatomical connection by injection of polysynaptic tracer pseudorabies virus (PRV)-152 into the gastric antrum and subsequent retrograde uptake into the Sst-GABA neurons. Lewin et al (2016) also established a functional connection using electrophysiological evidence. Their experiment used transgenic Sst-IRES-Cre mice expressing channelrhodopsin-2v (ChR2-tdTomato, a red fluorescent tracer), archaerhodopsin-3/enhanced green fluorescent protein (EGFP) (ArchT-EGFP), and GCAMP3 (a fusion of GFP, calmodulin, and myosin peptide sequence M13). These reporter genes enabled optogenetic stimulation and allowed for identification of Cre-altered somatostatin neurons via fluorescence. When Lewin *et al* (2016) stimulated coronal brain sections containing the brainstem nuclei of the nucleus tractus solitarius (NTS) and DMV with light, whole-cell voltage and current clamp recordings revealed that activation of Sst-GABA neurons led to an increase in IPSCs and the inhibition of DMV output neurons projecting to the gastric antrum.

### *Inhibition of Sst-GABA Neurons by Anorexigenic Agonists*

Previous studies have established the role of melanocortin agonists in the regulation of gastric-antrum activity through the DMV (Richardson *et al*, 2013). The aforementioned study by Lewin et al. (2016) also showed that Sst-GABA neurons are under tonic inhibition by melanocortin and mu-opioid agonists. As previously mentioned, activation of Sst-GABA neurons leads to inhibition of DMV output neurons. However, when pretreated with melanocortin and MOR agonists, Sst-GABA neurons demonstrate significant inhibition in the frequency of action potentials while DMV output neurons were no longer inhibited (Lewin et al., 2016).

Furthermore, other studies have shown that the melanocortin receptor ligands mediate gastrointestinal signals through vagal afferent fibers (De Jonghe, Hayes, & Bence, 2011; Wan et al., 2008), and that the brain-derived neurotrophic factor, a downstream effector of melanocortin signalling, also plays an important role in anorexigenic decreases in food intake (Bariohay, Lebrun, Moyse, & Jean, 2005). Overall, the above-cited researchers have established that Sst-GABA neurons exerts inhibitory regulation on gastric reflex through melanocortin and other anorexigenic agonists.

***Neuroanatomical Identification by Viral Tracing***

Although viral tracing has long been used to map neural circuitry, the pseudorabies virus (PRV) method developed by Pomeranz *et al.* (2017) serves as the most appropriate neuroanatomical tracer for the purposes of our experiment. Unlike its predecessors, PRV-152 does not leak to non-Cre-recombinase neurons as PRV-Bartha does (Pomeranz *et al*, 2017), nor is it limited to monosynaptic labelling of only primary neurons as adenovirus and adeno-associated virus are (Saunders & Sabatini, 2015). A member of the subfamily Alphaherpesvirinae, PRV is able to spread polysynaptically in the mammalian nervous system. The developed strain contains a PRV tk gene and a fluorescent reporter; to maintain a typically inactive status, a synthetic intron has been inserted into the PRV tk gene and one of the exons has been inverted with FLEx switch (Flip-Excision system). Only upon exposure to Cre recombinase would the inversion be flipped, resulting in a functional reporter gene. Thus, the PRV and reporter genes are only activated in the presence of Cre expression in certain cells of a transgenic mouse.

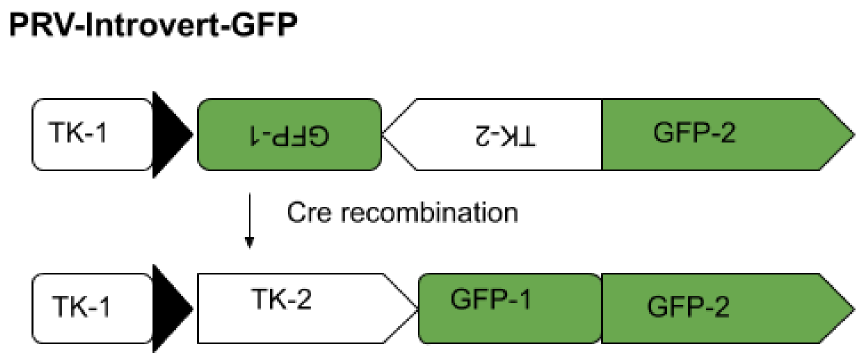


*Figure 1*. Cre-dependent PRV-Introvert-GFP mechanism. Under regular circumstances, the Cre-dependent PRV-Introvert has a reversed reporter gene, thus preventing it from labelling infected neurons. However, upon exposure to Cre-expressing neurons, the Cre-lox recombination will flip the inversion so that the neuron expresses EGFP. Only transgenic mice with Cre-expressing neurons will activate the viral tracer and show fluorescent labelling.

PRV successfully labelled targeted neurons in previous studies involving the vagal-gastric circuit. Herman, Gillis, Vicini, Dretchen, & Sahibzada (2012) used PRV in Sprague-Dawley rats to identify the NTS neurons projecting to the gastric antrum: Niedringhaus et al. (2008) used dual injection of PRV-151 into the lower esophageal sphincter and crural diaphragm to trace connections and points of convergence in the neural circuitry. Similarly, in the aforementioned experiment by Lewin et al. (2016), where the neuroanatomical tracer helped to establish that Sst-GABA neurons in the DMV were indeed anatomically connected to the gastric antrum. Upon injection of PRV-152 EGFP into the gastric-antrum of Sst-IRES-Cre mice expressing tdTomato, subsequent labelling of Sst-GABA neurons upstream showed connection via the vagal circuitry. By imaging harvested brain slices under a confocal microscope, we can count the labelled neurons on FIJI and overlay them onto camera lucida drawings to determine the infected brain regions (Pomeranz *et al*, 2017).

In our present study, we aimed to expand upon previous research by applying the polysynaptic retrograde tracer PRV-152 to determine sources of anorexigenic agonists upstream from the DMV and NTS.

## Model Systems

Adhering to the model described by Lewin *et al* (2016), we used transgenic mice with Cre expression in either the Sst neurons. Specifically, we worked with Sst-Cre mice ($Sst^{tm2.1(cre)Zjh}$/J; Jackson lab #013044; The Jackson Laboratory, Bar Harbor, ME US) (Taniguchi et al, 2011). We worked with both male and female subjects of postnatal day 17-21 (Lewin *et al.* 2016). Our lab maintained the mice under conditions described by Richardson *et al* (2013).

For the neuroanatomical tracer, we used viral strain PRV-152 (gifted by Dr. Lisa Pomeranz; Pomeranz *et al*, 2017).

## Hypothesis and Rationale

In previous studies, researchers have suggested functional connections between the stomach, DMV circuit, and certain parts of the brain such as the amygdala (Pomrenze *et al*, 2015), hypothalamus (Biag *et al*, 2012), and raphe origin (van der Kooy, Koda, McGinty, Gerfen, & Bloom, 1984) due to their role in gastric motility and energy balance.

We anticipate verifying these anatomical and functional connections by showing fluorescent labelling of these regions through activation of the Cre-dependent PRV-Introvert-EGFP polysynaptic tracer.

## Section II: Materials and Methods

*Ethical Approval*

Mice were killed by rapid decapitation in accordance with the National Institutes of Health (Bethesda, MD, USA) guidelines and with the approval of the Georgetown University Animal Care and Use Committal. All procedures were approved by the Institutional Review Board.

**Injection of Retrograde Tracer**

We injected the PRV-152 neuroanatomical tracer (gifted by Dr. Lisa Pomeranz; Pomeranz *et al*, 2017) into the mouse antrum. We used transgenic mice with Cre-recombinase expression in Sst interneurons ($Sst^{tm2.1(cre)Zjh}$/J; Jackson lab #013044; The Jackson Laboratory, Bar Harbor, ME US). We worked with both male and female mice of postnatal day 17-21 (Lewin *et al.* 2016).

Surgical procedures followed those described by Richardson *et al* (2013) for applying the DiI tracer into the stomach. Instead of DiI, we injected 5 $\mu$L of PRV-152. All other aspects of the surgery remained identical.

We then harvested the brains of these mice one to three days post PRV injection using the method described by Lewin *et al*, 2016. Mice were rapidly decapitated and perfused before we made 250 $\mu$m coronal and horizontal slices of the brain with a vibratome (VT1000S; Leica Microsystems, Wetzlar, Germany). In the time between PRV injection and decapitation, retrograde uptake of PRV would spread the fluorescent tracer polysynaptically from the stomach into the vagal-gastric circuitry. Upon exposure to Cre recombinase in the Sst interneurons, the Cre-dependent EGFP reporter gene in the virus would activate and leave fluorescent labelling on their upstream synaptic projections (Pomeranz *et al*, 2017).

**Image Analysis**

We performed neuroanatomical analyses of the EGFP labelling in harvested brain slices using ImageJ. We referenced Franklin and Paxinos's (2008) atlas of the mouse brain to determine the regions of the brain that project to Sst neurons (Niedringhaus *et al*, 2008). We took confocal images of the brain slices at 20X magnification for fluorescence and binocular images for slice anatomy. In ImageJ, we used the multi-point tool to count out fluorescently labelled neurons on a max-intensity Z-Project of the stack of images, cross-referencing with the original stack to ensure we did not double-count cells. We then saved the XY-coordinates of the point series through the ROI manager and exported the dotted image to Krita, an image editing software. Through Krita, we overlaid the brain slices and labelled neurons onto a slice of the mouse atlas. For hindbrain slices, we overlaid fluorescent labels onto coronal Slice 93. For forebrain slices, we overlaid fluorescent labels onto horizontal slice 151.

A

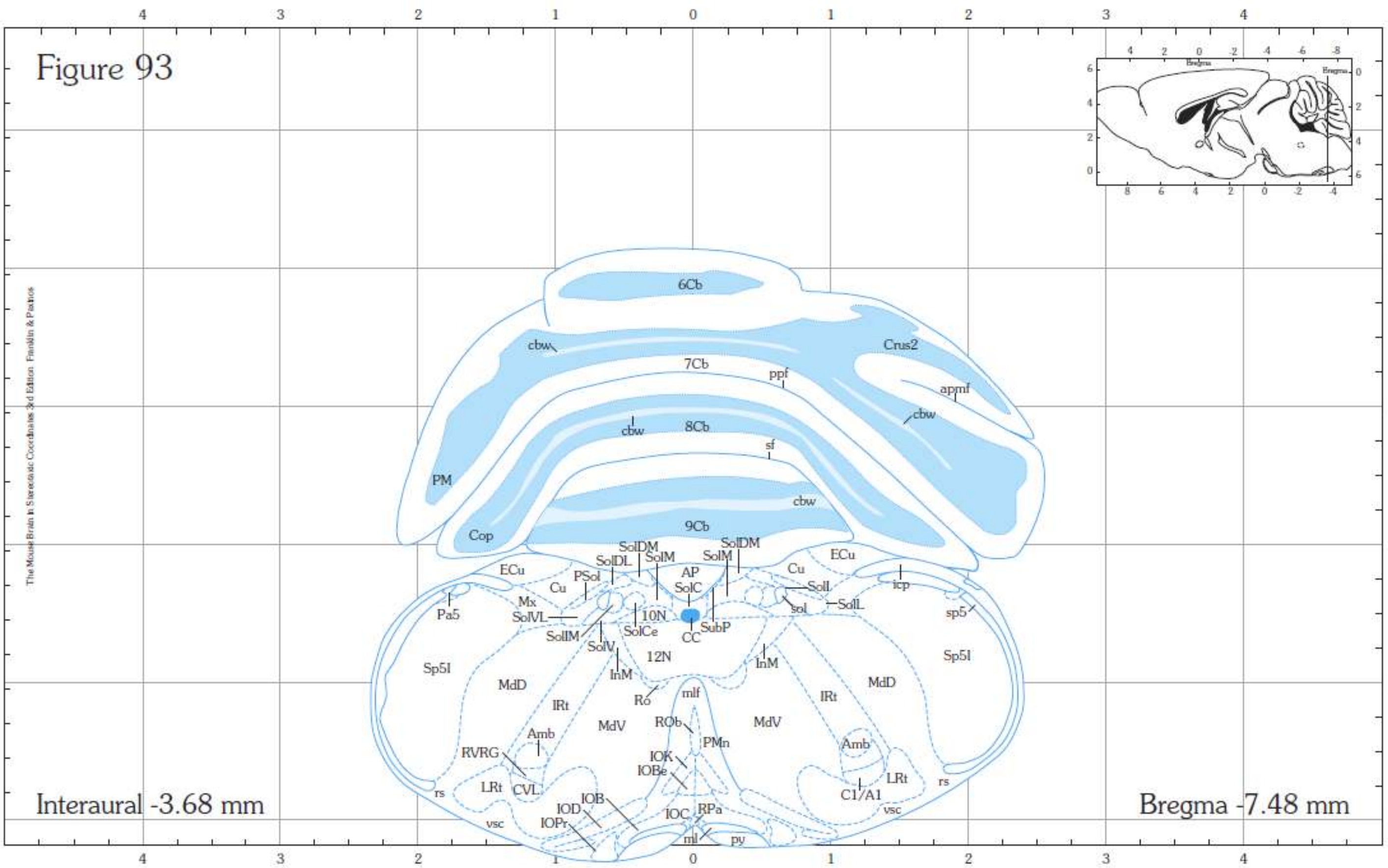
Figure 93
The Mouse Brain in Stereotaxic Coordinates 3rd Edition Franklin & Paxinos
6Cb
cbw
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cbw
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Cu
PSol
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SolIM
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SolCe
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MdV
ROb
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IOK
IOBe
LRt
CVL
C1/A1
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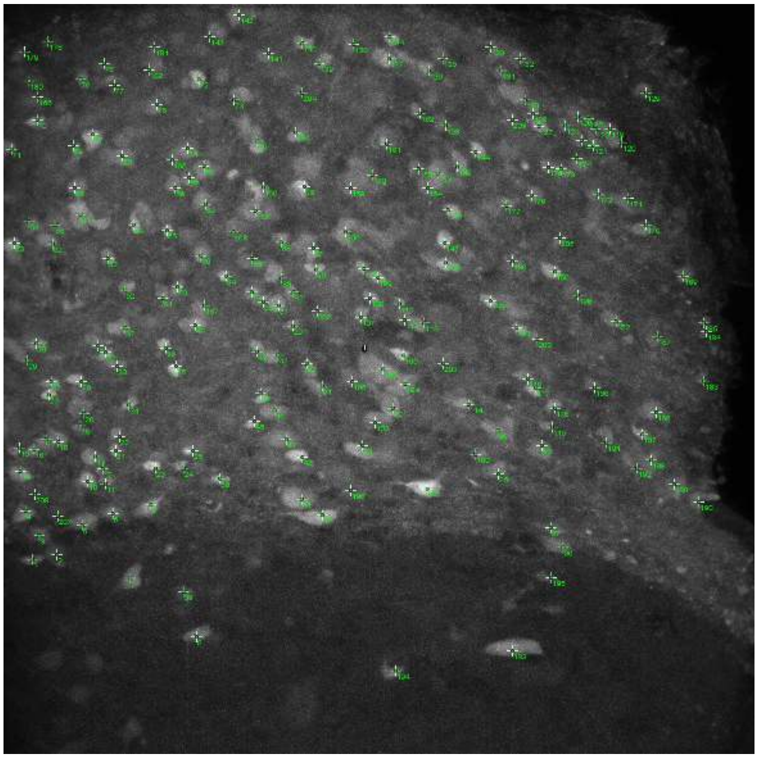
IOB
IOD
IOPr
IOC
RPa
vsc
ml
py
Interaural -3.68 mm
Bregma -7.48 mm


B

*Figure 2.* Anatomical reference for the hindbrain (Franklin & Paxinos, 2008). A) Based on the approximate stereotaxic coordinates of the coronal brain slices, we determined the reference of interest to be coronal Slice 93. The predominant feature of our area of interest is in the DMV, denoted by “10N.” B) We cross-referenced with the Nissl-stained plate of Slice 93 to ensure our approximate coordinates were correct.



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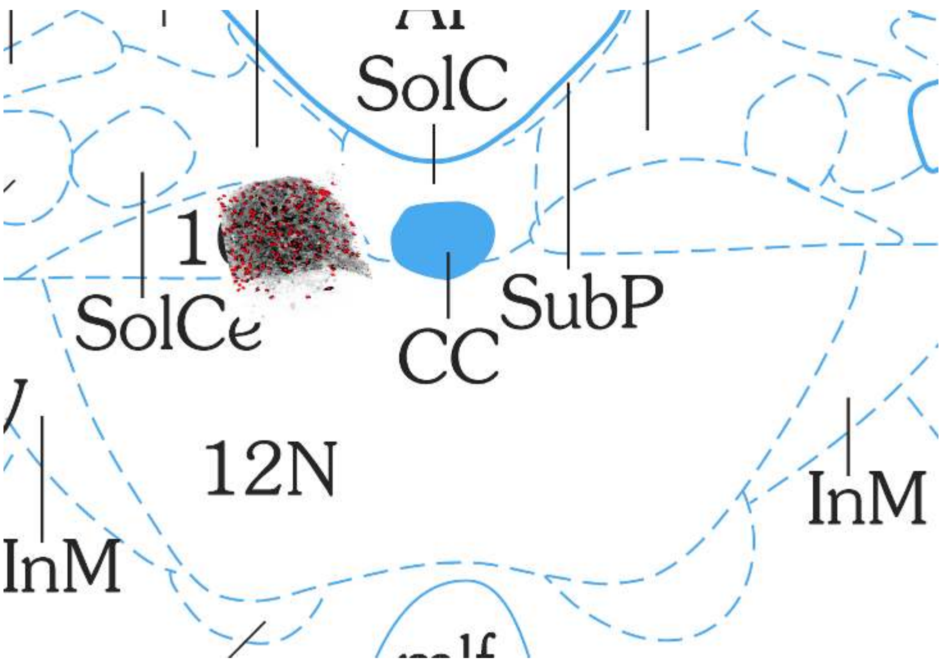


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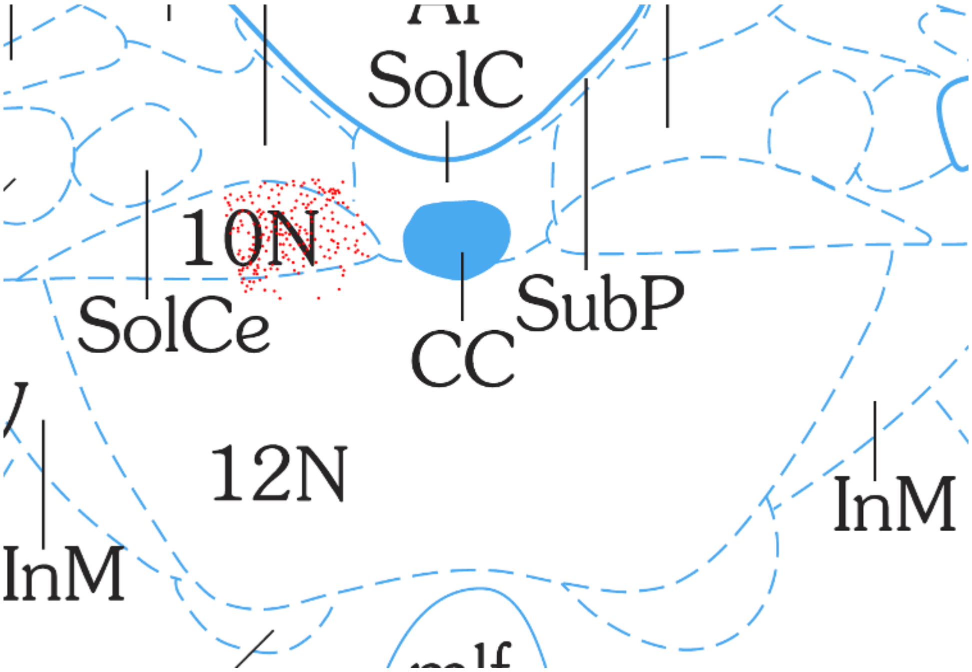


C

*Figure 3*. Representative image overlay procedure for the 20X confocal image of a hindbrain slice. A) In ImageJ, we counted fluorescently labelled neurons with the multi-point tool and exported the flattened max-intensity Z-Project. B) In Krita, we went over the points with a constant-radius hard brush in a separate layer and added filter masks to the max-intensity Z-Project to increase contrast and invert black and white. These image renderings allowed for more a more accurate overlay of the fluorescent neurons onto the corresponding DMV region of Slice 93 of the mouse brain atlas. Magnification of Slice 93 can be determined by referencing stereotaxic coordinates from Figure 1. C) After alignment, we left only the constant-radius dots to represent the locations of labelled neurons.

A

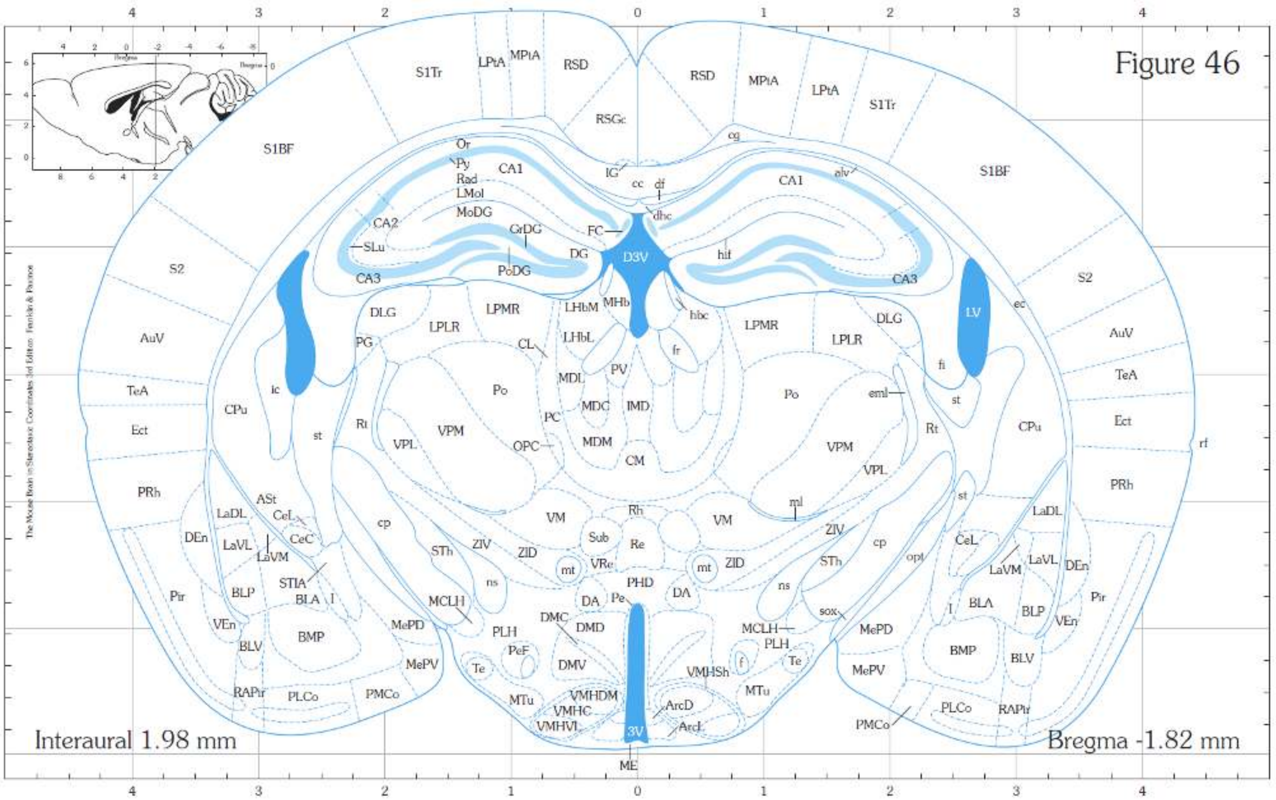
Figure 46
Interaural 1.98 mm
Bregma -1.82 mm
The Mouse Brain in Stereotaxic Coordinates 3rd Edition Franklin & Paxinos
S1Tr
LPtA
MPtA
RSD
RSGc
S1BF
cg
Or
Py
Rad
LMol
MoDG
CA1
CA2
CA3
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PoDG
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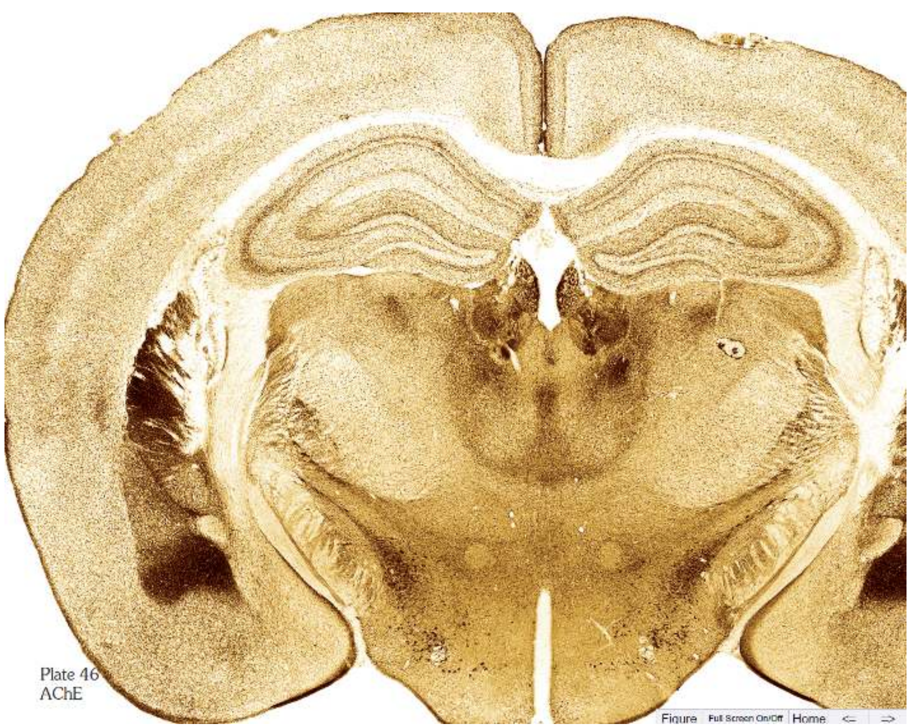
Plate 46
AChE
Figure
Full Screen On/Off
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B

*Figure 4.* Anatomical reference for the forebrain (Franklin & Paxinos, 2008). A) Based on the approximate stereotaxic coordinates of the coronal brain slices, we determined the reference of interest to be coronal Slice 46. B) We cross-referenced with the AChE-stained plate of Slice 46 to ensure our approximate coordinates were correct.

**Safety considerations**

As an unlicensed minor, I am not able to perform any of the euthanization or surgeries mentioned in this experimental design. I only worked with images and image analysis, so I was not exposed to any procedures that may warrant safety considerations.

## Section III: Results

**Confocal verification of PRV-152 functionality with Sst-Cre mice**

We confirmed the functionality of the PRV-152 tracer analyzing slices for fluorescence under a confocal microscope (Figure 5). Upon contact Cre-expressing Sst neurons in the DMV or NTS of the transgenic mice, the Cre-lox system should activate the EGFP reporter gene in the viral tracer. Here, we could see successful PRV-induced fluorescence in both the hindbrain area and the forebrain area.

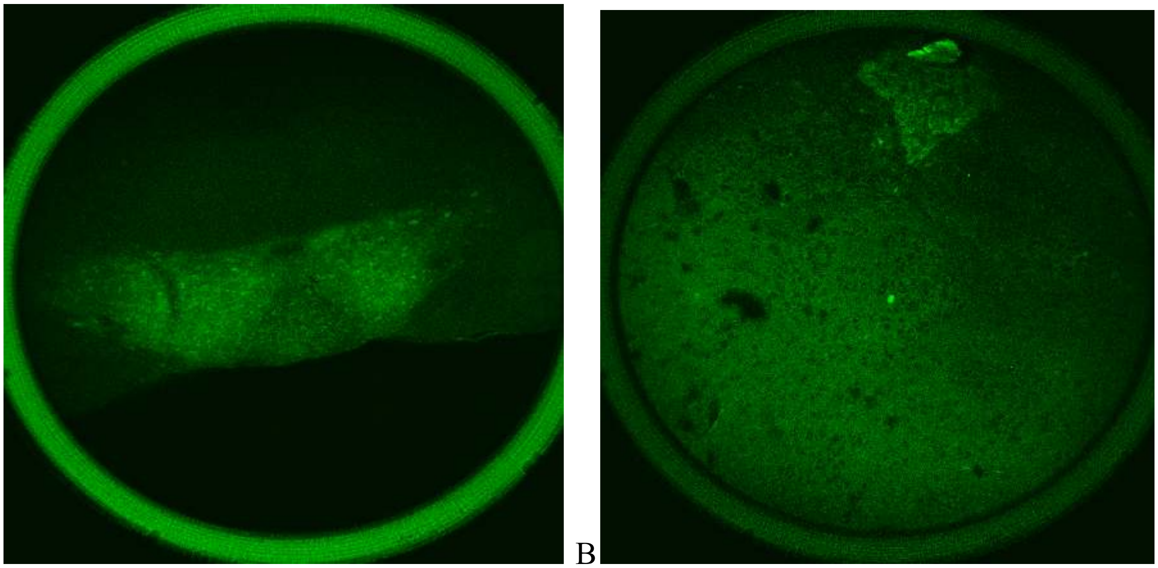
A B

*Figure 5*. Verification of activated PRV-152 fluorescence. To give a preliminary check on the successful activation of PRV-152 upon exposure to Cre-expressing Sst neurons, we analyzed fluorescence under a confocal microscope. A) shows a coronal hindbrain slice at 10X magnification. B) shows a coronal forebrain slice at 20X magnification.

**Control identification of Sst-Cre in the DMV and NTS**

From the hindbrain slices, we overlaid fluorescent neurons onto coronal slices of the Paxinos and Franklin mouse brain Atlas. We used images taken under the bifocal microscope as reference for the anatomy of each slice. Based on the opening of the fourth ventricle, we placed most overlays on Slice 93 and one on Slice 91 (Figure 6).

In each case, we observed fluorescent neurons in the areas we expected: the dorsal motor nucleus of the vagus and the nucleus solitary tract. These placements confirmed the activation of the PRV-152 reporter gene upon exposure to Cre-expressing Sst and Npy in the two regions.

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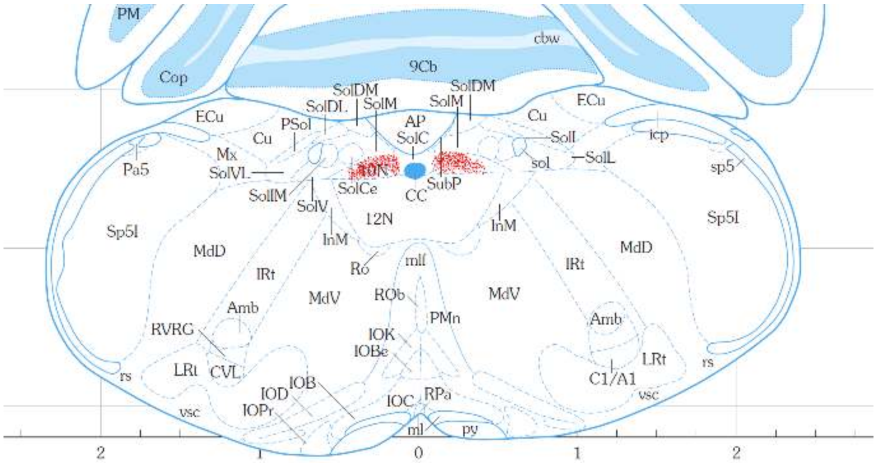
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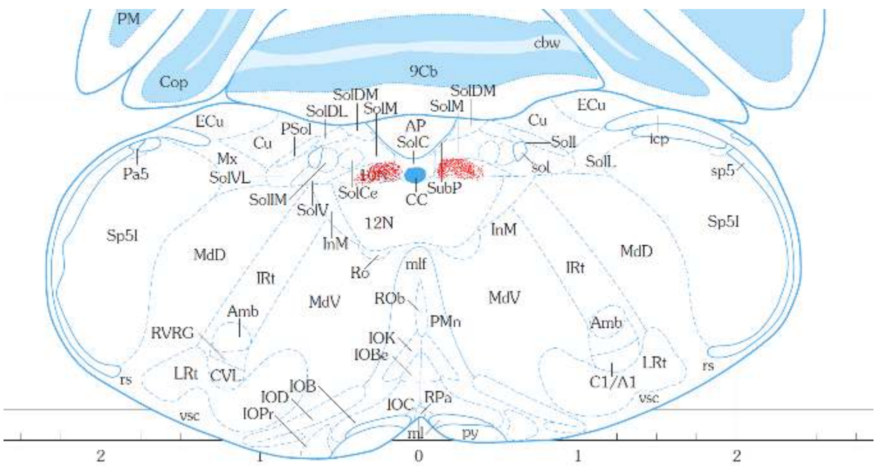
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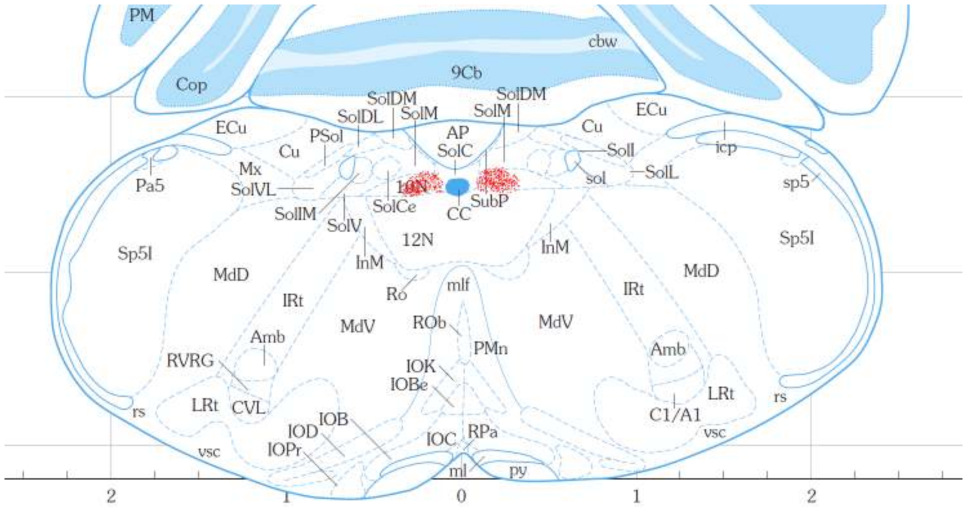
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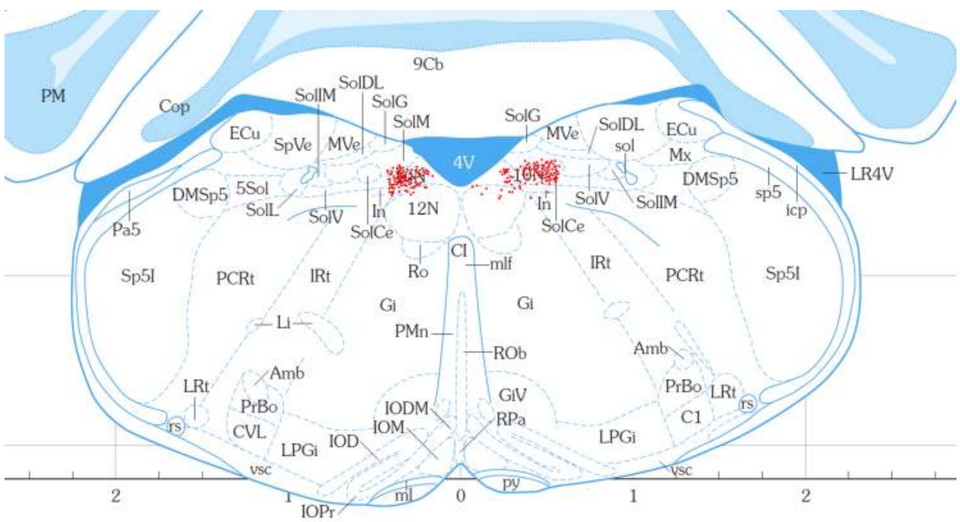
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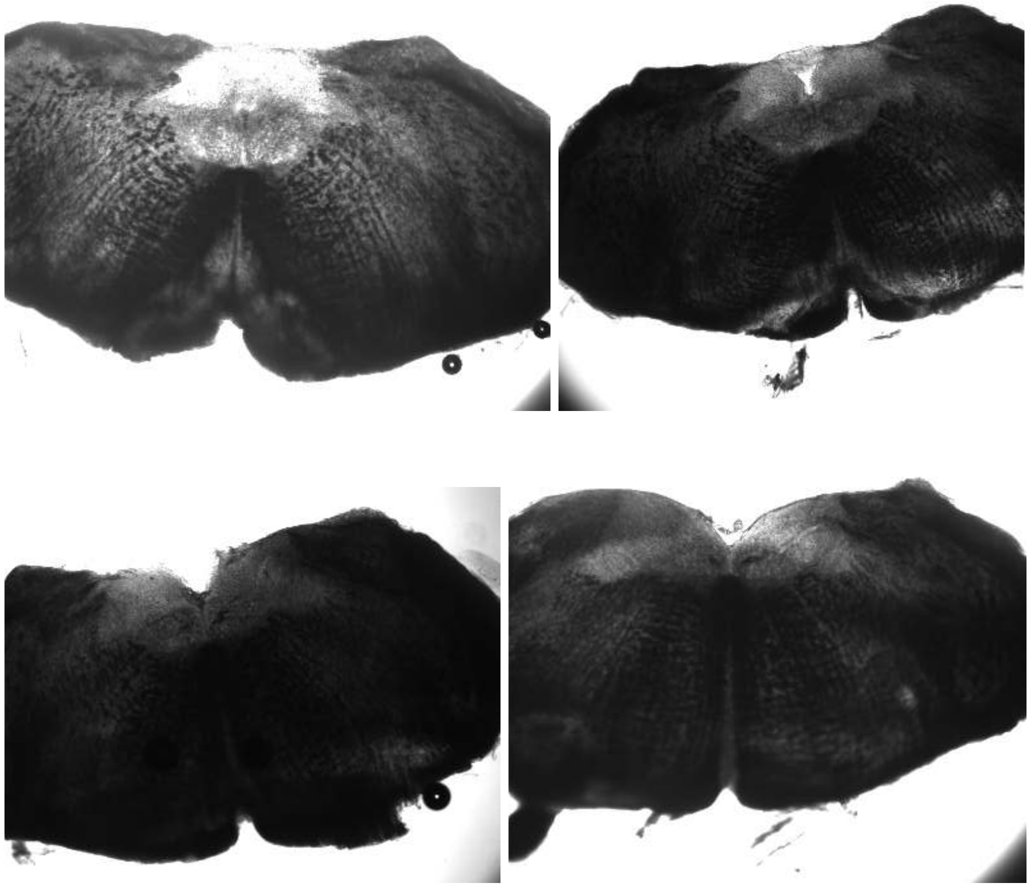

*Figure 6*. Analysis of hindbrain fluorescence. A) Representative overlay of fluorescently traced neurons from hindbrain slices over Slice 93 or Slice 91 of Paxinos & Franklin's (2008) brain atlas. Determination of which slice (more caudal or more rostral) depended on the anatomy of corresponding images taken under the bifocal microscope (B, 2X magnification). No matter the slice, we observed the presence of fluorescent Sst neurons in the DMV and NTS.

**Identification of upstream regions of Sst in the forebrain**

From the compiled sections, we overlaid fluorescent neurons onto Slice 43 of Paxinos and Franklin's mouse brain atlas (Figure 7). This figure represents a coronal slice in the forebrain, at stereotaxic coordinates of Interaural: 1.98 mm and Bregma: -1.82 mm. Notably, we observed fluorescence in the areas of the hypothalamus: dorsomedial hypothalamic nucleus, arcuate nucleus of hypothalamus, paraventricular thalamic nucleus, ventromedial thalamic nucleus, ventral reuniens thalamic nucleus, and submedius thalamic nucleus. Dots appeared in greatest density in the arcuate nucleus of hypothalamus.

However, it should be noted that the overlay seen in Figure X1 does not represent all the collected data. We had six slices that yielded successful fluorescence, but due to the confocal images lacking any sort of neuroanatomical landmarks, we could not overlay these neurons onto Figure X1 with confidence of their coordinates. However, by referencing the corresponding images taken under a bifocal microscope, we can infer that these fluorescent neurons are from the amygdala.

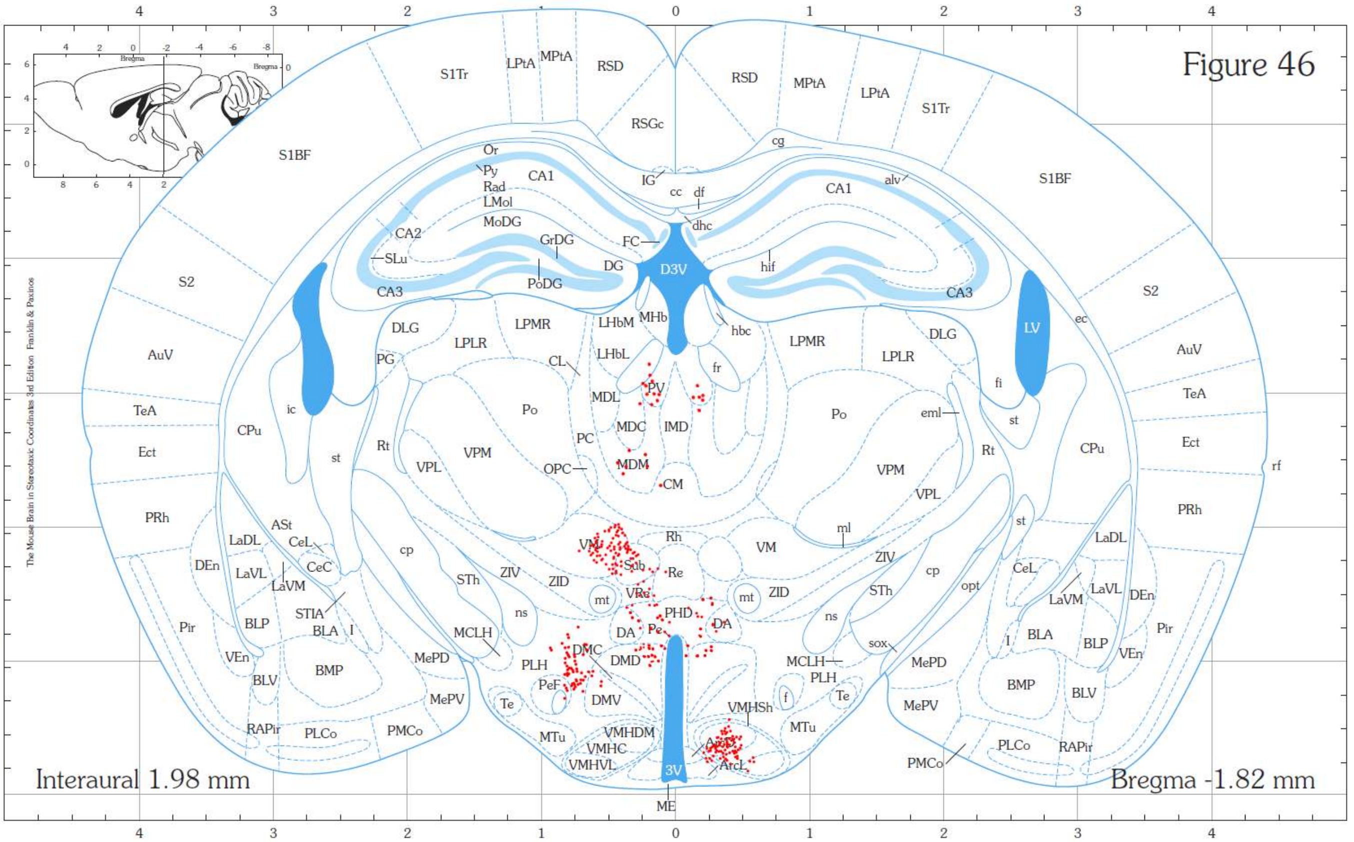


*Figure 7*. Analysis of forebrain fluorescence. We compiled the fluorescent neurons from seven different slices and overlaid them onto Slice 43 of Paxinos and Franklin's brain atlas (2008).

## Section IV: Discussion

In this study, we used PRV-152 to polysynaptically trace the upstream regions to the DMV and NTS through retrograde action. From analyses of neural fluorescence under the confocal microscope, we determined three major afferent areas of the brain: the hypothalamus, the raphe origin, and the amygdala.

We confirmed the functionality of the PRV-152 tracer by first analyzing hindbrain regions of the DMV and NTS. From previous studies (Lewin *et al*, 2016), we had already confirmed both the anatomical and function connection of the DMV and NTS to the antrum through the vagus nerve using PRV-152 and electrophysiological stimulation. Since both the DMV and NTS showed fluorescence consistently through all coronal slices, we could be reasonably certain that the Cre-expressing Sst neurons had successfully activated the fluorescent reporter gene in the PRV-152.

Upstream from the DMV and NTS, we found the greatest certainty in the anatomical and functional connections between the hypothalamus and the DMV and NTS. The overlaid fluorescent neurons were present in many sub-sections of the hypothalamus, including the well-cited paraventricular thalamic nucleus. These results corroborate previous findings that showed a regulatory connection between the hypothalamus and weight homeostasis (Lee, 2015). The hypothalamus has also been shown to contain cholecystokinin (CCK), a neurotransmitter present in the brain-gut axis and acts as a modulator of gastrointestinal function through the vagus nerve (Little, Horowitz, & Feinle-Bisset, 2005). CCK has been shown to be involved in the inhibition of gastric emptying of the stomach and food intake, which supports the role of the hypothalamus in the innervation of the antrum. There has also been electrophysiological evidence to support the role of the hypothalamus. Through a connection with the cerebellum, the hypothalamus may regulate food intake through projections of critical feeding signals, including the aforementioned CCK (Zhu & Wang, 2007).

Furthermore, previous studies also support the role of the paraventricular thalamus nucleus and arcuate nucleus of hypothalamus (Van Bockstaele, Peoples, & Telegan, 1999).

However, Van Bockstaele *et al* (1999) focused on finding monosynaptic relations, whereas our use of the PRV-152 allowed for polysynaptic tracing from the antrum to the hypothalamus. Other studies also showed signal transmission from the paraventricular thalamus nucleus to the vagal-vago circuit through the NTS. Not only does this connection contribute to gastric function, but also supports the role of the hypothalamus in the hypothalamic–pituitary–adrenal axis and in chronic depression (O'Keane, Dinan, Scott, & Corcoran, 2005).

In addition to the hypothalamic neurons, we also saw fluorescence in the amygdala. Although confocal images of these slices did not reveal obvious anatomical landmarks to place the neurons in an exact region of the amygdala, the images from the bifocal microscope confirmed the locale of these neurons. Therefore, we could reasonably conclude the anatomical connection between the amygdala and antrum, though specific regions could not be identified.

Previous studies support our findings as researchers have shown that stimulation of the vagus nerve would send signals to the amygdala and hypothalamus through the NTS (George *et al*, 2000). Another lab showed that electrical stimulation in the amygdala led to inhibition of both NTS and DMV neurons (Zhang, Cui, Tan, Jiang, & Fogel, 2003). Zhang *et al* (2003) also demonstrated anatomical connections between the amygdala and DMV using retrograde tracer Alexa-Fluor-555-conjugated cholera toxin subunit B. Cholera toxin subunit B is a monosynaptic tracer, so we were able to expand on their results by using the polysynaptic PRV-152.

Functionally, the role of the amygdala in this upstream circuit makes sense as it has been previously shown to regulate feeding behavior through the NTS (Giraudo, Kotz, Billington, & Levin, 1998). Electrical stimulation of the amygdala has also shown inhibitory regulation on gastric motility (Liubashina, Bagaev, & Khotiantsev, 2002). Furthermore, by comparing vagotomized rats with or without stimulation at the amygdala, researchers have shown that the amygdala plays an important role in gastric function by stimulating gastric acid secretion through the vagus nerve (Kim *et al*, 1990).

Despite previous evidence of an anatomical and functional connection between the Raphe nucleus and the antrum (Richardson *et al*, 2013), we could not definitively conclude the role of the Raphe origin in this study. Due to the number of slices lacking neuroanatomical landmarks and the ambiguity of the bifocal images (Figure 8), it is possible that evidence of anatomical connection between the Raphe origin and the antrum was present, but not confirmed with confidence. In either case, our lack of definitive findings does not suggest against potential future evidence of this connection.

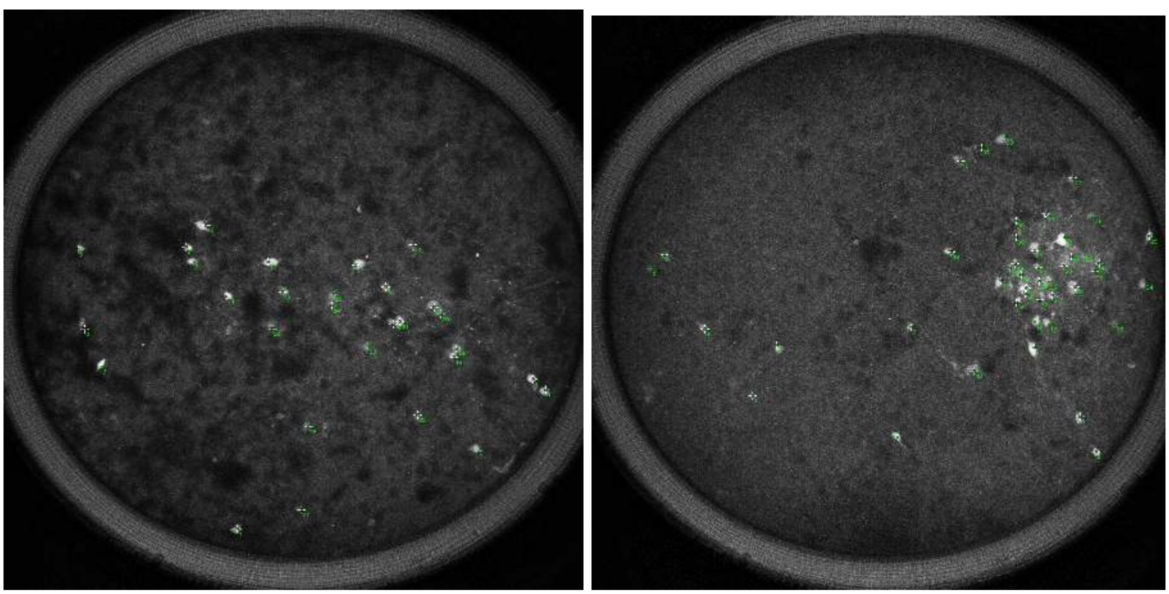

*Figure 8*. Representative slices that lacked anatomical landmarks. Due to several confocal images lacking clear anatomical landmarks, we could not overlay these fluorescent neurons onto the brain atlas with confidence in their exact placement.

## Section V: Conclusions

Our findings largely supported our hypothesis based on previous research. In particular, we observed clear fluorescent traces of PRV-152 in the hypothalamus, confirming the neuroanatomical connection between the hypothalamus and the antrum through the vagal circuit. Furthermore, fluorescence in the amygdala corroborated previous studies establishing the role of the amygdala in gastric function using polysynaptic tracers and electrophysiological stimulation.

In addition to the corroboration of previously established relations, we also observed connections with the dorsomedial hypothalamic nucleus, ventromedial thalamic nucleus, ventral reuniens thalamic nucleus, and submedius thalamic nucleus. In the future, we may hope to establish and support these findings with electrophysiological evidence of the functional connection.

However, we were not able to conclude a connection to the Raphe origin with confidence, as the lack of anatomical markers in our confocal and bifocal images did not offer sufficient clarity for the placement of fluorescent neurons in some slices. In the future, we recommend improving the imaging methodology by providing landmarks to ensure accurate identification of the location of each slice.

Furthermore, we hope to expand our experiment to include investigations on the connections with Npy neurons. By employing dual retrograde tracers, we can co-localize the circuitry of Sst and Npy neurons to better understand their role in regulating gastric function and determine whether signalling cascades from the medial nucleus of the solitary tract (mNTS) converge at Sst or Npy neurons in the DMV. We can also expand our study to include more subjects so that we can harvest brains at different time periods. By varying the time of post-injection harvest, we would better understand the pathways between the antrum and the melanocorgenic circuit.

## Section VI: Cited Literature

Bariohay, B., Lebrun, B., Moyse, E., & Jean, A., (2005). Brain-Derived Neurotrophic Factor Plays a Role as an Anorexigenic Factor in the Dorsal Vagal Complex, *Endocrinology*, *146(12),* 5612–5620. https://doi.org/10.1210/en.2005-0419

Biag, J., Huang, Y., Gou, L., Hintiryan, H., Askarinam, A., Hahn, J. D., … Dong, H. W. (2012). Cyto- and chemoarchitecture of the hypothalamic paraventricular nucleus in the C57BL/6J male mouse: a study of immunostaining and multiple fluorescent tract tracing. *J Comp Neuro.*, 520(1), 6–33. doi:10.1002/cne.22698

Chan, Y. S., Ko, J. K., & Cho, C.H. (1995). Role of dorsal motor nucleus of vagus in gastric function and mucosal damage induced by ethanol in rats. *Dig. Dis. Sci., 40(11)*, 2312-2316. doi: 10.1007/bf02063230

Cruz, M. T., Dezfuli, G., Murphy, E. C., Vicini, S., Sahibzada, N., & Gillis, R. A. (2019). GABAB Receptor Signaling in the Dorsal Motor Nucleus of the Vagus Stimulates Gastric Motility via a Cholinergic Pathway. *Front. Neurosci*, *13*, 967. doi:10.3389/fnins.2019.00967

De Jonghe, B. C., Hayes, M. R., & Bence, K. K. (2011). Melanocortin control of energy balance: evidence from rodent models. *CMLS, 68*(15), 2569-2588. doi:10.1007/s00018-011-0707-5

Franklin, K. & Paxinos, G. (2008). *The Mouse Brain in Stereotaxic Coordinates, Third Edition*. Cambridge: Academic Press.

George, M. S., Sackeim, H. A., Rush, A. J., Marangell, L.B., Nahas, Z, Husain, M. M., et al. (2000). Vagus nerve stimulation: a new tool for brain research and therapy. *Biol Psychiatry*, 47:287–95. doi:10.1016/S0006-3223(99)00308-X

Giraudo, S.Q., Kotz, C.M., Billington, C.J., & Levin, A.S. (1998). Association between the amygdala and nucleus of the solitary tract in mu-opioid induced feeding in the rat. *Brain Res, 802*(1-2), 184-188. DOI: 10.1016/s0006-8993(98)00602-7

Herman, M. A., Gillis, R. A., Vicini, S., Dretchen, K. L., & Sahibzada, N. (2012). Tonic GABAA receptor conductance in medial subnucleus of the tractus solitarius neurons is inhibited by activation of μ-opioid receptors. *J. Neurophysiology, 107*(3), 1022-1031. doi:10.1152/jn.00853.2011

Lee, H. S., Kim, M. A., & Waterhouse B. D, (2005) Retrograde double-labeling study of common afferent projections to the dorsal raphe and the nuclear core of the locus coeruleus in the rat. *J Comp Neurol, 481*(2), 179-193. doi: 10.1002/cne.20365

Lewin, A. E., Vicini, S., Richardson, J., Dretchen, K. L., Gillis, R. A., & Sahibzada, N. (2016). Optogenetic and pharmacological evidence that somatostatin-GABA neurons are important regulators of parasympathetic outflow to the stomach. *J Physiol, 594*(10), 2661-2679. doi:10.1113/JP272069

Little, T. J., Horowitz, M., & Feinle-Bisset, C. (2005). Role of cholecystokinin in appetite control and body weight regulation. *Obes Rev., 6*(4), 297–306. doi:10.1111/j.1467-789X.2005.00212.x

Liubashina, O, Bagaev, V., & Khotiantsev, S. (2002). Amygdalofugal modulation of the vago-vagal gastric motor reflex in rat. *Neuroscience Letters*, *325*(3), 183-186. doi:10.1016/S0304-3940(02)00289-6

Kim, M.S., Jo, Y.H., Yoon, S.H, Hahn, S.J., Rhie, D.J., Chung, C.K., & Choi, H. (1990). Electrical stimulation of the medial amygdala facilitates gastric acid secretion in conscious rats. *Brain Research, 524*(2), 208-212. doi: 10.1016/0006-8993(90)90692-5

Niedringhaus, M., Jackson, P. G., Pearson, R., Shi, M., Dretchen, K., Gillis, R. A., & Sahibzada, N. (2008). Brainstem sites controlling the lower esophageal sphincter and crural diaphragm in the ferret: a neuroanatomical study. *Auton Neurosci, 144*(1-2), 50-60. doi:10.1016/j.autneu.2008.09.007

O'Keane, V., Dinan, T. G., Scott, L., & Corcoran, C. (2005). Changes in hypothalamic–pituitary–adrenal axis measures after vagus nerve stimulation therapy in chronic depression. *Biol Psychiatry*, 58:963–8. doi:10.1016/j.biopsych.2005.04.049

Pomeranz, L. E., Ekstrand, M. I., Latcha, K. N., Smith, G. A., Enquist, L. W., & Friedman, J. M. (2017). Gene Expression Profiling with Cre-Conditional Pseudorabies Virus Reveals a Subset of Midbrain Neurons That Participate in Reward Circuitry. *Neurosci, 37*(15), 4128-4144. doi:10.1523/JNEUROSCI.3193-16.2017

Pomrenze, M. B., Millan, E. Z., Hopf, F. W., Keiflin, R., Maiya, R., Blasio, A., … Messing, R. O. (2015). A Transgenic Rat for Investigating the Anatomy and Function of Corticotrophin Releasing Factor Circuits. *Front. Neurosci*, 9, 487. doi:10.3389/fnins.2015.00487

Richardson, J., Cruz, M. T., Majumdar, U., Lewin, A., Kingsbury, K. A., Dezfuli, G., . . . Sahibzada, N. (2013). Melanocortin signaling in the brainstem influences vagal outflow to the stomach. *J Neurosci, 33*(33), 13286-13299. doi:10.1523/JNEUROSCI.0780-13.2013

Taniguchi, H., He, M., Wu, P., Kim, S., Paik, R., Sugino, K., … Huang, Z. J. (2011). A resource of Cre driver lines for genetic targeting of GABAergic neurons in cerebral cortex. *Neuron*, 71(6), 995–1013. doi:10.1016/j.neuron.2011.07.026

Saunders, A., Sabatini, B. (2015). Cre Activated and Inactivated Recombinant Adeno-Associated Viral Vectors for Neuronal Anatomical Tracing or Activity Manipulation. *Curr. Protoc. Neurosci, 72:1.24.1-1.24.15.* doi: 10.1002/0471142301.ns0124s72

van der Kooy, D., Koda, L.Y., McGinty, J.F., Gerfen, C.R. and Bloom, F.E. (1984), The organization of projections from the cortes, amygdala, and hypothalamus to the nucleus of the solitary tract in rat. *J. Comp. Neurol*., 224: 1-24. doi:10.1002/cne.902240102

Wan, S., Browning, K. N., Coleman, F. H., Sutton, G., Zheng, H., Butler, A., … Travagli, R. A. (2008). Presynaptic melanocortin-4 receptors on vagal afferent fibers modulate the excitability of rat nucleus tractus solitarius neurons. *J. Neuroscience*, *28*(19), 4957–4966. doi:10.1523/JNEUROSCI.5398-07.2008

Zhang, X., Cui, J., Tan, Z., Jiang, C., & Fogel, R. (2003). The central nucleus of the amygdala modulates gut-related neurons in the dorsal vagal complex in rats. *J. Physiol*, 553(3), 1005-1018. doi: 10.1113/jphysiol.2003.045906

Zhu, J. & Wang, J. (2008). The Cerebellum in Feeding Control: Possible Function and Mechanism. *Cell Mol Neurobiol*, 28(4), 469–478. doi:10.1007/s10571-007-9236-z